128

```
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$ ./ns3 shell
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$ python3 examples/tutorial/first.py
At time +2s client sent 128 bytes to 10.1.1.2 port 9
At time +2.00225s server received 128 bytes from 10.1.1.1 port 49153
At time +2.00225s server sent 128 bytes to 10.1.1.1 port 49153
At time +2.00451s client received 128 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
Tx Bytes:  156
Rx Bytes:  156
Tx Packets:  1
Rx Packets:  1
Lost Packets:  0
Mean Delay:  0.0022528
Throughput:  69.33333333333333
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
Tx Bytes:  156
Rx Bytes:  156
Tx Packets:  1
Rx Packets:  1
Lost Packets:  0
Mean Delay:  0.0022528
Throughput:  69.33333333333333
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$
```

256

```
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$ ./ns3 shell
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$ ^C
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$ python3 examples/tutorial/first.py
At time +2s client sent 256 bytes to 10.1.1.2 port 9
At time +2.00246s server received 256 bytes from 10.1.1.1 port 49153
At time +2.00246s server sent 256 bytes to 10.1.1.1 port 49153
At time +2.00492s client received 256 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
Tx Bytes:  284
Rx Bytes:  284
Tx Packets:  1
Rx Packets:  1
Lost Packets:  0
Mean Delay:  0.0024576
Throughput:  126.22222222222223
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
Tx Bytes:  284
Rx Bytes:  284
Tx Packets:  1
Rx Packets:  1
Lost Packets:  0
Mean Delay:  0.0024576
Throughput:  126.22222222222223
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$
```

512

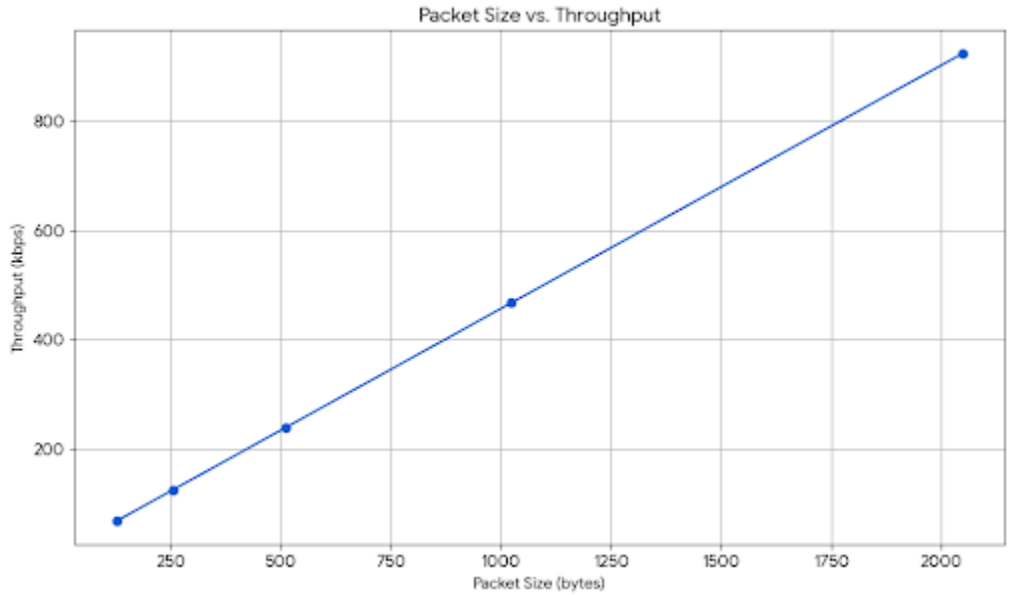
```
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$ ./ns3 shell
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$ python3 examples/tutorial/first.py
At time +2s client sent 512 bytes to 10.1.1.2 port 9
At time +2.00287s server received 512 bytes from 10.1.1.1 port 49153
At time +2.00287s server sent 512 bytes to 10.1.1.1 port 49153
At time +2.00573s client received 512 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
Tx Bytes:  540
Rx Bytes:  540
Tx Packets:  1
Rx Packets:  1
Lost Packets:  0
Mean Delay:  0.0028672
Throughput:  240.0
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
Tx Bytes:  540
Rx Bytes:  540
Tx Packets:  1
Rx Packets:  1
Lost Packets:  0
Mean Delay:  0.0028672
Throughput:  240.0
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$
```

1024

```
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$ ./ns3 shell
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$ python3 examples/tutorial/first.py
At time +2s client sent 1024 bytes to 10.1.1.2 port 9
At time +2.00369s server received 1024 bytes from 10.1.1.1 port 49153
At time +2.00369s server sent 1024 bytes to 10.1.1.1 port 49153
At time +2.00737s client received 1024 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
Tx Bytes:  1052
Rx Bytes:  1052
Tx Packets:  1
Rx Packets:  1
Lost Packets:  0
Mean Delay:  0.0036864
Throughput:  467.55555555555554
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
Tx Bytes:  1052
Rx Bytes:  1052
Tx Packets:  1
Rx Packets:  1
Lost Packets:  0
Mean Delay:  0.0036864
Throughput:  467.55555555555554
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$
```

2048

```
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$ ./ns3 shell
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$ python3 examples/tutorial/first.py
At time +2s client sent 2048 bytes to 10.1.1.2 port 9
At time +2.00536s server received 2048 bytes from 10.1.1.1 port 49153
At time +2.00536s server sent 2048 bytes to 10.1.1.1 port 49153
At time +2.01072s client received 2048 bytes from 10.1.1.2 port 9
FlowID: 1 (UDP 10.1.1.1/49153 --> 10.1.1.2/9)
Tx Bytes:  2076
Rx Bytes:  2076
Tx Packets:  1
Rx Packets:  1
Lost Packets:  0
Mean Delay:  0.00536
Throughput:  922.6666666666666
FlowID: 2 (UDP 10.1.1.2/9 --> 10.1.1.1/49153)
Tx Bytes:  2076
Rx Bytes:  2076
Tx Packets:  1
Rx Packets:  1
Lost Packets:  0
Mean Delay:  0.00536
Throughput:  922.6666666666666
ubuntu@ubuntu-HP-280-Pro-G8-Microtower-PC:~/Downloads/ns-allinone-3.40/ns-3.40$
```



The graph clearly shows that **throughput increases as packet size increases.**

This happens because of **fixed overhead**. Every packet sent must include headers (like IP and UDP headers), which have a fixed size.

- **With small packets,** these headers make up a large portion of the total data, causing a lot of waste and low efficiency.
- **With large packets,** the same fixed-size headers represent a tiny fraction of the total data. This is much more efficient, as the network spends most of its time sending your actual data (payload) instead of just headers.